
CS772 Project Report

Dhruv Bansal (220359)

IIT-Kanpur

Department of Aerospace Engineering

dhrubb22@iitk.ac.in

Himanshu Gupta (220453)

IIT-Kanpur

Department of Civil Engineering

ghimanshu22@iitk.ac.in

Abstract

Training machine learning models to be robust to out-of-distribution (OOD) samples remains a key challenge in safe deployment. While prior works have explored leveraging auxiliary outlier datasets or synthesizing outliers in latent space, generating semantically plausible and diverse photo-realistic outliers remains an open problem. In this work, we propose a probabilistic framework for OOD data generation and detection. Our approach replaces von Mises-Fisher modeling of the latent space with a richer and uncertainty-aware formulation using Projected Normal Distributions (PND). By modeling each class as a PND on the unit hypersphere, we capture both directional alignment and intra-class variance. This allows us to sample embeddings from low-likelihood regions more precisely, representing high-confidence OOD samples. These sampled embeddings are then used to condition a text-guided diffusion model, enabling synthesis of photo-realistic and semantically controllable OOD images.

Our code is available at. <https://github.com/dhruv-ban1/PND-OOD>

1 Introduction

Outlier detection in a dataset during model training is required since the outliers significantly affect the performance of a machine learning model. In the case of classification issues out of Distribution (OOD), detection is extremely crucial since new data not specified but similar to something could be misclassified by the models and subsequently impact the performance. Since machine learning models are being deployed more and more in real-world settings, the detection and proper handling of OOD inputs has become a critical requirement.

We have taken the same approach as suggested in (1), but we have introduced a new distributional method to build the text-conditioned latent space. The outliers are drawn from this space and then translated into pixel-space images with a Stable Diffusion model conditioned on the corresponding text conditions.

2 Background and Literature review

The paper “Dream the Impossible: Outlier Imagination with Diffusion Models” introduces a framework called **DREAM-OOD**, which proposes a novel approach to synthesize photo-realistic outliers using diffusion models for enhancing out-of-distribution (OOD) detection in machine learning. The motivation lies in the challenge of generating effective outliers when only in-distribution (ID) data is available, as real-world applications often face unexpected inputs beyond training distributions.

The approach begins by learning a text-conditioned latent space from the ID dataset. Given training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n \sim P_{XY}$, the goal is to train a classifier $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^C$ that is accurate on ID data and can also detect when a sample x is drawn from an unknown class $y \notin \mathcal{Y}$. The diffusion model used is text-conditioned and produces images from text embeddings $z_y = T(y)$, where T is a

fixed text encoder like that from CLIP. Normally, image generation with diffusion models follows $x \sim P(x|z_y)$. However, textual descriptions for outliers are not straightforward to construct, as by definition, outliers represent unknown or unspecified semantic concepts.

To address this, DREAM-OOD introduces a method to train an image encoder $h_\theta : \mathcal{X} \rightarrow \mathbb{R}^m$, such that the image embeddings $z = h_\theta(x)$ align with their corresponding class text embeddings $T(y)$. This alignment is encouraged using a temperature-scaled softmax loss:

$$\mathcal{L} = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[-\log \frac{\exp(T(y)^\top z/t)}{\sum_{j=1}^C \exp(T(y_j)^\top z/t)} \right], \quad z = \frac{h_\theta(x)}{\|h_\theta(x)\|_2}. \quad (1)$$

This loss encourages the image features to cluster around their class centroids on the hypersphere, effectively approximating a von Mises-Fisher distribution:

$$p_m(z; \mu_c, \kappa) = Z_m(\kappa) \exp(\kappa \mu_c^\top z), \quad P(y = c|z) = \frac{\exp(\mu_c^\top z/t)}{\sum_{j=1}^C \exp(\mu_j^\top z/t)}, \quad (2)$$

where μ_c is the centroid corresponding to class c , $\kappa = 1/t$, and $Z_m(\kappa)$ is a normalization constant.

Once this latent space is learned, the next step is to synthesize outliers. The idea is to sample embeddings v that lie in the low-likelihood regions of the learned latent space, i.e., far from the ID embeddings. This is done non-parametrically: the k-nearest neighbor (k-NN) distance of each embedding z' with respect to the dataset $Z = \{z_1, \dots, z_n\}$ is calculated as:

$$d_k(z', Z) = \|z' - z^{(k)}\|_2, \quad (3)$$

where $z^{(k)}$ is the k -th nearest neighbor. Embeddings with high k-NN distance are considered on the boundary of the ID distribution. Around these boundary points, outlier embeddings $v \sim \mathcal{N}(z_i, \sigma^2 I)$ are sampled, and the one with the largest k-NN distance is selected.

The sampled embedding v is then used in place of a class token $T(y)$ in the diffusion model, giving:

$$x_{\text{ood}} \sim P(x|v), \quad (4)$$

which results in a pixel-space image that is semantically distinct from ID images. Thus, DREAM-OOD allows generating realistic, inspectable outliers.

These synthetic outliers are used to regularize the classifier during training using an additional loss $\mathcal{L}_{\text{ood}}$ defined by:

$$\mathcal{L}_{\text{ood}} = \mathbb{E}_{x_{\text{ood}}} \left[-\log \frac{1}{1 + \exp(\phi(E(f_\theta(x_{\text{ood}}))))} \right] \quad (5)$$

$$+ \mathbb{E}_{x \sim P_{\text{in}}} \left[-\log \frac{\exp(\phi(E(f_\theta(x))))}{1 + \exp(\phi(E(f_\theta(x))))} \right], \quad (6)$$

where ϕ is a nonlinear MLP, $E(f(x))$ is the energy function, and $f_\theta(x)$ is the classifier's logit. The full training objective combines standard classification with this regularization: $\mathcal{L}_{\text{CE}} + \beta \cdot \mathcal{L}_{\text{ood}}$, where β balances the two.

Extensive experiments show that training with DREAM-OOD outliers improves OOD detection across various benchmarks (e.g., ImageNet-100, CIFAR-100). Metrics such as AUROC and FPR95 confirm that DREAM-OOD outperforms traditional methods like ODIN, Mahalanobis distance, KNN scoring, and even other generative approaches like GANs and latent-space sampling (VOS, NPOS).

Moreover, DREAM-OOD can be adapted for ID synthesis. By selecting points in the latent space with low k-NN distances and sampling close to them, one can generate inlier images, which the paper calls **DREAM-ID**. These synthetic inliers are also useful for enhancing generalization. Tests on ImageNet variants (ImageNet-A, ImageNet-V2) demonstrate that DREAM-ID offers better generalization than standard augmentations (e.g., CutMix, AutoAugment).

In conclusion, DREAM-OOD provides a principled method for generating meaningful, photo-realistic outliers without requiring explicit textual prompts for unknown classes. The theoretical formulation via vMF distributions and the practical algorithmic pipeline—latent alignment, low-likelihood sampling, diffusion-based decoding, and classification regularization—work together to enhance the robustness of machine learning models in uncertain environments.

3 Our Methodology

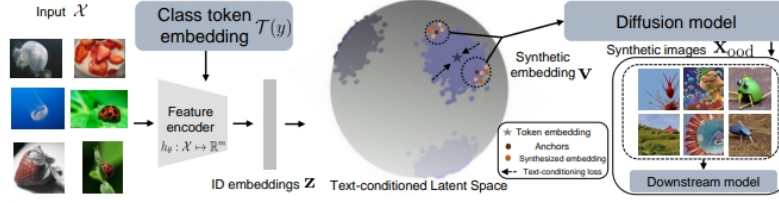


Figure 1: Illustration of outlier generation workflow (1)

The original DREAM-OOD framework proposes a generative approach for generating semantically meaningful outliers by aligning L2-normalized image embeddings with class token embeddings on a hypersphere using the von Mises-Fisher (vMF) distribution. Although effective, this approach treats all within-class embeddings as concentrated around a single mean direction with fixed concentration which ignores the variability and uncertainty.

In contrast, we propose a probabilistic latent modeling framework based on the **Projected Normal Distribution (PND)**. Instead of modeling the embedding space purely via directional cosine similarity (as in vMF), we treat each class as a distribution over the sphere induced by projecting a Gaussian distribution in $\mathbb{R}^d$ onto $\mathbb{S}^{d-1}$. This allows us to model both the *mean direction* and the *spread* (covariance) of embeddings for each class.

This probabilistic formulation enables:

- Uncertainty-aware latent space modeling
- Anisotropic and data-driven cluster shapes
- Outlier detection based on likelihood under class-conditional distributions

By replacing cosine-based softmax scoring with PND-based likelihoods, we obtain more compact and informative latent clusters, which improves both discriminative performance and the quality of generated outliers.

3.1 Projected Normal Distribution (PND)

Let $X \sim \mathcal{N}(\mu, \Sigma)$ be a multivariate normal distribution in $\mathbb{R}^d$. The Projected Normal Distribution (PND) arises from the transformation:

$$z = \frac{X}{\|X\|}, \quad z \in \mathbb{S}^{d-1}$$

The resulting density of z on the unit hypersphere is referred to as the Projected Normal Distribution:

$$z \sim \text{PN}(\mu, \Sigma)$$

The exact density function of the PND is defined as:

$$p(z) = \int_0^\infty r^{d-1} \mathcal{N}(rz; \mu, \Sigma) dr$$

where rz is a radial transformation of the unit vector z scaled by $r \in \mathbb{R}_+$.

This expression integrates over all rays extending from the origin in the direction of z , evaluating the likelihood under the base Gaussian at each point. While this is mathematically well-defined, the integral does not have a closed-form solution in general, making direct evaluation and sampling computationally intractable for high-dimensional spaces.

Therefore, in practice, we use approximate or unnormalized forms of the log-density:

$$\log p(z | \mu, \Sigma) \approx -\frac{1}{2} \log(z^\top \Sigma^{-1} z) + \frac{(z^\top \mu)^2}{2z^\top \Sigma z}$$

This approximation retains the key geometric properties of the distribution:

- The first term penalizes directions with high inverse covariance
- The second term encourages alignment with the mean μ , scaled by local uncertainty

This formulation allows efficient training and likelihood-based filtering in high-dimensional spherical latent spaces.

Although the true density of the PND is intractable, an approximate unnormalized log-likelihood on the hypersphere can be written as:

$$\log p(z \mid \mu, \Sigma) = -\frac{1}{2} \log(z^\top \Sigma^{-1} z) + \frac{(z^\top \mu)^2}{2z^\top \Sigma z}$$

This formulation captures both alignment with the mean direction and dispersion shaped by Σ .

3.2 PND-based Multiclass Training Objective

Let $z_i = \frac{h_\theta(x_i)}{\|h_\theta(x_i)\|}$ be the L2-normalized embedding of an input image x_i , and let y_i denote its class label. For each class $j \in \{1, \dots, C\}$, we model the class conditional distribution as a PND with parameters (μ_j, Σ_j) .

We define a PND-based log-likelihood score for each class j :

$$s_j(z_i) = -\frac{1}{2} \log(z_i^\top \Sigma_j^{-1} z_i) + \frac{(z_i^\top \mu_j)^2}{2z_i^\top \Sigma_j z_i}$$

The multiclass PND training loss is then defined via a softmax over class log-likelihoods:

$$\mathcal{L}_{\text{PND-softmax}} = -\log \frac{\exp(s_{y_i}(z_i))}{\sum_{j=1}^C \exp(s_j(z_i))}$$

Substituting the PND score, we obtain:

$$\mathcal{L}_{\text{PND-softmax}} = -\log \frac{\exp \left[-\frac{1}{2} \log(z_i^\top \Sigma_{y_i}^{-1} z_i) + \frac{(z_i^\top \mu_{y_i})^2}{2z_i^\top \Sigma_{y_i} z_i} \right]}{\sum_{j=1}^C \exp \left[-\frac{1}{2} \log(z_i^\top \Sigma_j^{-1} z_i) + \frac{(z_i^\top \mu_j)^2}{2z_i^\top \Sigma_j z_i} \right]}$$

This loss generalizes the vMF softmax loss by incorporating variance-aware, probabilistic similarity via PND. During training, the encoder h_θ is optimized to maximize the likelihood of each sample under its class-specific PND.

3.3 Sampling Outliers from the Latent Space

A key component of our framework is the ability to generate plausible but semantically out-of-distribution samples by exploring the learned latent space. While prior works such as DREAM-OOD use heuristic boundary sampling (e.g., k-nearest neighbors) to approximate low-density regions, our approach provides a probabilistic formulation rooted in the Projected Normal Distribution (PND).

After training the encoder h_θ using the PND-based loss (Section ??), we model the distribution of image embeddings for each class c as:

$$z \sim \text{PN}(\mu_c, \Sigma_c), \quad z \in \mathbb{S}^{d-1}$$

where μ_c and Σ_c are the empirical class mean and covariance estimated from ID embeddings:

$$\mu_c = \mathbb{E}[z \mid y = c], \quad \Sigma_c = \text{Cov}[z \mid y = c]$$

3.3.1 Low-Likelihood Outlier Sampling

To generate outlier embeddings, we sample from the underlying multivariate normal:

$$x \sim \mathcal{N}(\mu_c, \Sigma_c), \quad \tilde{z} = \frac{x}{\|x\|}$$

Algorithm 1 PND-OD: Outlier Imagination with Projected Normal Modeling and Diffusion

Input: In-distribution training data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, initial model parameters θ , diffusion model P_{diff} .

Output: Synthetic outlier images $\mathcal{X}_{\text{OOD}}$

Phases: Phase 1: Learning the PND-Aligned Latent Space. Phase 2: Outlier Sampling and Diffusion-based Generation.

while Phase 1 do

- Extract token embeddings μ_y and estimate class-wise covariance Σ_y for each class label $y \in \mathcal{Y}$.
- Learn the normalized latent representation space $z_i = \frac{h_\theta(x_i)}{\|h_\theta(x_i)\|}$ by minimizing the PND-based contrastive loss

while Phase 2 do

- For each class y , sample latent vector $x \sim \mathcal{N}(\mu_y, \Sigma_y)$ from low likelihood .
 - Normalize: $\tilde{z} = \frac{x}{\|x\|}$.
 - Retain $\tilde{z}$ if $\log p(\tilde{z} \mid \mu_y, \Sigma_y) < \tau$.
 - Decode $\tilde{z}$ into the pixel-space image x_{OOD} by using diffusion model
-

Then we evaluate the approximate log-likelihood of each sample under its generating class PND:

$$\log p(\tilde{z} \mid \mu_c, \Sigma_c) = -\frac{1}{2} \log(\tilde{z}^\top \Sigma_c^{-1} \tilde{z}) + \frac{(\tilde{z}^\top \mu_c)^2}{2\tilde{z}^\top \Sigma_c \tilde{z}}$$

Samples with likelihoods below a threshold τ are retained as outliers:

$$\mathcal{Z}_{\text{OOD}} = \{\tilde{z} \mid \log p(\tilde{z} \mid \mu_c, \Sigma_c) < \tau\}$$

3.3.2 Guided Decoding via Diffusion

Each sampled outlier $\tilde{z} \in \mathcal{Z}_{\text{OOD}}$ is projected to match the dimension of the CLIP text encoder output, and used to replace the class-conditioned text embedding in a pretrained Stable Diffusion model:

$$x_{\text{outlier}} \sim P_{\text{diffusion}}(x \mid \tilde{z})$$

This process allows us to decode latent outlier embeddings into photo-realistic images that are semantically close to, yet statistically distinct from, the ID distribution.

4 Results and Discussion

To assess the effectiveness of the Projected Normal Distribution (PND) in modeling class-conditional latent spaces, we visualize the learned embeddings using t-SNE in Figure 2. For comparison, we also include the latent representation produced by the von Mises-Fisher (vMF) distribution from the original DREAM paper in Figure 3.

The t-SNE plot of our approach (Figure 2) demonstrates highly structured and tightly clustered class-wise features. Each cluster forms a radially spread yet directionally consistent pattern around the origin, indicating that the PND effectively captures directional variance in the data. Notably, the inter-class separation is significant, and each cluster maintains a distinct elliptical spread, implying that the model preserves both angular diversity and intra-class compactness. This emergent structure aligns well with the geometric properties of PND, which models the projection of a multivariate normal vector onto the unit hypersphere, offering greater flexibility in capturing the spread and concentration of class-specific features.

In contrast, the baseline vMF embeddings (Figure 3) exhibit tighter but somewhat overlapping clusters, with class centers being relatively uniformly spread. While the vMF’s concentration parameter κ enforces directional clustering, it limits expressiveness in capturing intra-class variability. The PND overcomes this by enabling elliptical or anisotropic shapes in the latent space due to its underlying Gaussian construction prior to projection.

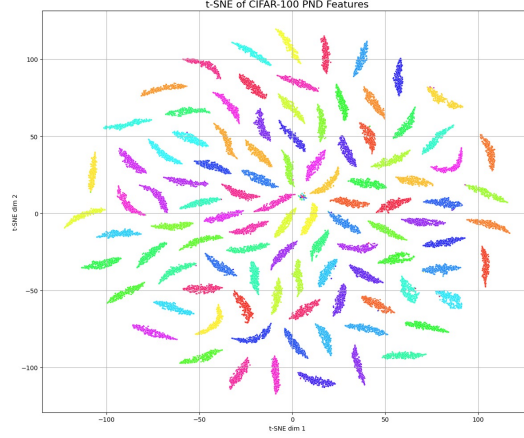


Figure 2: t-SNE of CIFAR-100 features learned using Projected Normal Distribution.

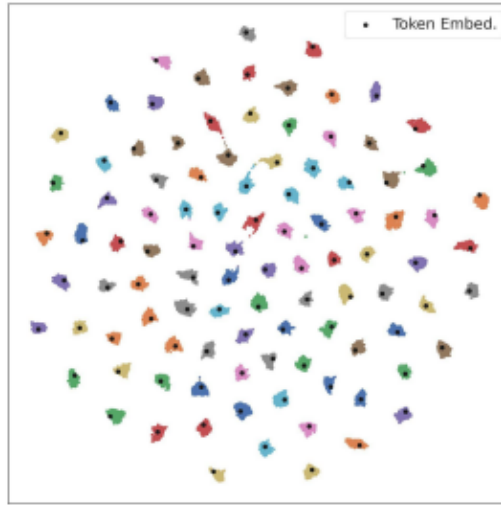


Figure 3: Latent space from DREAM paper using von Mises-Fisher distribution.

Overall, the PND-based latent representation shows enhanced expressiveness, better inter-class separability, and controlled intra-class variability. These characteristics make it a strong alternative to vMF for applications that benefit from probabilistically grounded and geometrically interpretable hyperspherical embeddings, especially in OOD detection where marginal likelihood and angular diversity are critical.

5 Tools Used

For Our project, we have used several open-source tools and libraries for modeling and experimentation. We use **PyTorch** as the primary deep learning framework to implement the encoder network and training with the Projected Normal Distribution (PND) loss. For image generation, we utilize the **Stable Diffusion** pipeline provided by the `diffusers` library from Hugging Face. The `transformers` library is used to access pretrained CLIP models for text and image embeddings. Additionally, NumPy, Matplotlib, and `scikit-learn` support data handling, visualization, and evaluation. Experiments are conducted in **Google Colab** and local GPU machines with CUDA acceleration.

6 Individual Contribution

This project was jointly carried out by both of us, with contributions in theoretical formulation, coding, experimentation, and documentation. Together, we explored the working of the existing DREAM-OOD framework and collaboratively proposed the use of Projected Normal Distributions (PND) as an alternative for modeling the latent space. One of us primarily led the coding and implementation tasks, including developing the training pipeline, integrating PND, handling sampling strategies, and generating OOD images using Stable Diffusion.

Meanwhile, the other focused more on the mathematical foundations, deriving the PND-based loss functions, likelihood evaluation, and outlier sampling strategy.

Along with that, most of the components—such as finalizing the loss formulation, setting up experiments, and writing the report—were developed by both of us. This collaborative workflow ensured that both members gained hands-on experience with both theory and implementation, resulting in a well-rounded and jointly developed project.

7 Learning from Project

Through this project, we gained valuable insights into out-of-distribution (OOD) detection and how synthetic data generation can enhance the robustness of machine learning models. By thoroughly studying the original DREAM-OOD paper, we learned how the authors approached modeling latent spaces with directional embeddings and generating realistic outlier images using diffusion models. This exploration helped us appreciate the concept of how a model can "imagine" data that seems realistic but diverges from the training distribution, offering a deeper understanding of model generalization and robustness.

While investigating other methods for OOD generation, we came across the Projected Normal Distribution (PND) as an alternative to the original approach. PND provides a more flexible and uncertainty-aware framework for clustering data in a latent space, particularly when trying to form compact class clusters on a hypersphere. This new technique gave us exposure to probabilistic clustering, which differs significantly from traditional methods like k-means or cosine-based approaches, and increases our understanding of modern probabilistic modeling.

On the technical side, we gained hands-on experience with powerful tools such as ResNet for image feature extraction, CLIP for text-image alignment, and Stable Diffusion for generating images. We applied these tools in various stages of our pipeline, from image generation to embedding manipulation. Additionally, we honed our skills in writing mathematical formulations, building training pipelines, and utilizing libraries like PyTorch and Hugging Face. These experiences allowed us to connect theoretical concepts with practical applications, boosting our confidence in tackling similar deep learning challenges in the future.

In the context of Latent Diffusion Models (LDM), we learned how image generation can be steered by a text prompt, and how injecting an outlier embedding into the prompt can guide the model to produce out-of-distribution (OOD) images. The process begins by tokenizing a text prompt and converting it into embeddings using CLIP. Replacing one token with an outlier embedding shifts the model's understanding, causing the generated image to deviate from typical representations. This technique leverages the power of self-attention in the CLIP transformer and cross-attention in the U-Net model to align image features with the modified text embeddings, resulting in diverse and creative outputs that push the boundaries of the model's typical data distribution.

This project has broadened our understanding of generative models, latent space manipulation, and the generation of synthetic data that challenges the model's learned distributions, leading to a deeper appreciation of OOD detection and its practical implications.

8 Future Work

While our current work successfully demonstrates the generation of out-of-distribution (OOD) images by leveraging Projected Normal Distributions (PNDs) in the latent space and decoding them using a text-guided diffusion model, there remain several important directions for future development and expansion of this framework.

Learning a Classifier with Synthetic Outliers

One immediate extension of our work is the integration of the generated outlier images into the training pipeline of a downstream classifier. In the original DREAM-OOD paper, the authors not only generated semantically meaningful OOD samples but also demonstrated that augmenting a classifier’s training set with these synthetic outliers improved its OOD detection performance. Following this approach, we plan to construct hybrid datasets that include both in-distribution samples and our generated OOD samples, and train classifiers using confidence-based or outlier exposure loss functions. This can help in better decision boundary calibration and improve the classifier’s robustness to unseen or anomalous inputs.

Improving Embedding Quality and Diversity

Another promising future direction involves improving the diversity and expressiveness of outlier embeddings. Although PND modeling provides a probabilistically grounded approach to capture intra-class variance, the use of empirical class covariance matrices may not always yield optimal low-likelihood regions in high-dimensional latent spaces. Incorporating learnable priors, mixture models (e.g., PND-Mixture), or variational approximations of the class-wise distribution may offer improved sampling strategies. Furthermore, exploring different sampling schemes (e.g., rejection sampling, adversarial perturbations, or energy-based sampling) could lead to better control over outlier characteristics and increase sample diversity.

References

- [1] Xuefeng Du, Yiyu Sun, Xiaojin Zhu, and Yixuan Li. *Dream the Impossible: Outlier Imagination with Diffusion Models*. arXiv preprint arXiv:2309.13415, 2023. Available at: <https://arxiv.org/abs/2309.13415>.
- [2] Yifei Ming, Ying Fan, and Yixuan Li. POEM: out-of-distribution detection with posterior sampling. In Proceedings of the International Conference on Machine Learning, pages 15650–15665, 2022.
- [3] Daniel Hernandez-Stumpfhauser, F.JayBreidt† and Mark J. van der Woerd: The General Projected Normal Distribution of Arbitrary Dimension: Modeling and Bayesian Inference

Disclaimer

This project is an original work completed independently by us as part of the CS772 Course project. It is not associated with any previous course project, internship, or external organization. All ideas, experiments, and results presented in this report have been developed solely for this project and have not been submitted elsewhere.